Tutorial for Chapter 2 (Part 2)

Exercise 1.
You are given $\mathbb{P}(A \cup B) = 0.7$ and $\mathbb{P}(A \cup B^{c}) = 0.9$. Find $\mathbb{P}(A)$

Solution 1.
For this question draw a diagram to see that $A = (A \cup B) \setminus (A^{c} \cap B)$, hence

$$\mathbb{P}(A) = \mathbb{P}(A \cup B) - \mathbb{P}(B \cap A^{c}) = \mathbb{P}(A \cup B) - (1 - \mathbb{P}(A \cup B^{c})) = 0.7 - 0.1 = 0.6.$$

Exercise 2.
Suppose that A and B are mutually exclusive events for which $P(A) = .3$ and $P(B) = .5$, what is the probability that

(a) either A or B occurs?

(b) A occurs but B does not?

(c) both A and B occur?

Solution 2.
(a) either or implies the union,

$$p = P(A \cup B) = P(A) + P(B) = .8.$$

(b) Similarly,

$$p = P(A \cap B^c) = P(A) = .3.$$

(c) Write

$$P(A \cap B) = P(\emptyset) = 0.$$

Exercise 3.
Prove by induction that

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \sum_{i=1}^{n} P(E_i).$$

Solution 3.
Note that, for $n = 2$, from Theorem 4 of lecture notes 2, one gets

$$P(E_1 \cup E_2) \leq P(E_1) + P(E_2)$$

Then, assume that $P(E_1 \cup E_2 \cup \cdots \cup E_n) \leq \sum_{i=1}^{n} P(E_i)$ holds, one has

$$\begin{aligned} P\left(E_1 \cup E_2 \cup \cdots \cup E_n \cup E_{n+1}\right) &= P\left(\left(E_1 \cup E_2 \cup \cdots \cup E_n\right) \cup E_{n+1}\right) \\ &\leq P\left(E_1 \cup E_2 \cup \cdots \cup E_n\right) + P\left(E_{n+1}\right) \\ &\leq \sum_{i=1}^{n} P(E_i) + P(E_{n+1}) \\ &= \sum_{i=1}^{n+1} P(E_i), \end{aligned}$$

which completes the proof.

Exercise 4.

- There are 90 applicants for a job with the news department of a television station. Some of them are college graduates and some are not; some of them have at least three years' experience and some have not, with the exact breakdown being

	College graduates	Not college graduates
At least three years' experience	18	9
Less than three years' experience	36	27

- If the order in which the applicants are interviewed by the station manager is random, G is the event that the first applicant interviewed is a college graduate, and T is the event that the first applicant interviewed has at least three years' experience, determine each of the following probabilities directly from the entries and the row and column totals of the table:

  a. $P(G)$, $P\left(T^{\mathrm{c}}\right)$, $P\left(G \cap T\right)$

  b. $P\left(G^{\mathrm{c}} \cap T^{\mathrm{c}}\right)$, $P\left(T \mid G\right)$, $P\left(G^{\mathrm{c}} \mid T^{\mathrm{c}}\right)$

Solution 4.

a.

$$\begin{aligned} P(G) &= \frac{54}{90} = \frac{3}{5} \\ P(T^{\mathrm{c}}) &= \frac{63}{90} = \frac{7}{10} \\ P(G \cap T) &= \frac{18}{90} = \frac{1}{5} \end{aligned}$$

b.

$$P(G^c \cap T^c) = \frac{27}{90} = \frac{3}{10}$$
$$P(G \mid T) = \frac{18}{54} = \frac{1}{5}$$
$$P(G^c \mid T^c) = \frac{27}{63} = \frac{3}{7}$$

Exercise 5.
Toss a coin 3 times. Define:

- $H_1 = \{$Event that the first toss is head$\}$;
- $H_A = \{$the event that 2 of the 3 tosses heads$\}$.

Find the probability of H_1 given H_A.

Solution 5.
The sample space is $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

We have $H_1 = \{HHH, HHT, HTH, HTT\}$, $H_A = \{HHT, HTH, THH\}$, then $H_1 \cap H_A = \{HHH, HHT\}$. Thus the requested probability is given by

$$\begin{aligned} P\left(H_1 \mid H_A\right) &= \frac{P\left(H_1 \cap H_A\right)}{P(H_A)} \\ &= \frac{P\left[\{HHH, HHT\}\right]}{P(H_A)} \\ &= \frac{2/8}{3/8} = \frac{2}{3}. \end{aligned}$$

Exercise 6.
The probability of surviving a certain transplant operation is 0.55. If a patient survives the operation, the probability that his or her body will reject the transplant within a month is 0.20. What is the probability of surviving both of these critical stages?

Solution 6.
Define the events

- $S =$ "Surviving a certain transplant" $\Longrightarrow P(S) = 0.55$
- $R =$ " Body reject the transplant" $\Longrightarrow P(R) = 0.20$

The requested probability is given by

$$P(S \cap \bar{R}) = P(\bar{R} \mid S) \cdot P(S) = 0.8 \times 0.55 = 0.44$$

Note: To see that draw a tree diagram.

Exercise 7.
In a certain community, 8 percent of all adults over 50 have diabetes. If a health service in this community correctly diagnoses 95 percent of all persons with diabetes as having the disease and incorrectly diagnoses 2 percent of all persons without diabetes as having the disease, find the probabilities that

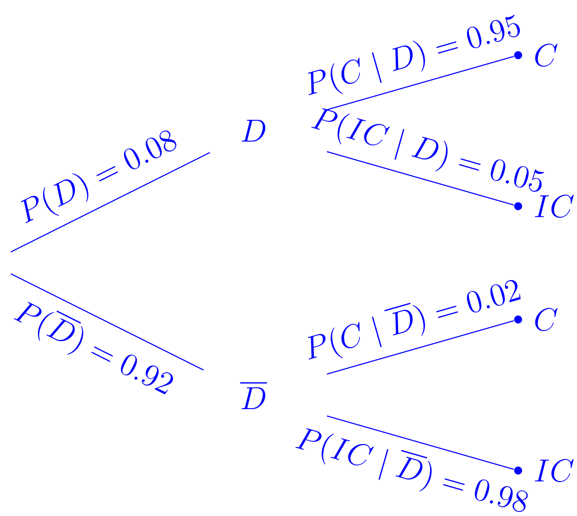
a. The community health service will diagnose an adult over 50 as having diabetes;

b. a person over 50 diagnosed by the health service as having diabetes actually has the disease.

Solution 7.
Define the events

- $D =$ " Adult over 50 have diabetes "
- $C =$ " Correct diagnose"
- $IC =$ "Incorrect diagnose"

Based on the tree diagram bellow the requested probability are



a. $P(C) = 0.08 \times 0.95 + 0.92 \times 0.02 = 0.0944$

b. Using bays theorem, we get

$$\begin{aligned} P(D \mid C) &= \frac{P(C \mid D)P(D)}{P(C)} \\ &= \frac{0.95 \times 0.08}{0.0944} = 0.805 \end{aligned}$$

Exercise 8.
Given that $P(A) = 0.2$, $P(B^c) = 0.6$, and $P(A \cap B) = 0.2$,
find

- $P(A \cup B)$
- $P(A \mid B)$
- $P(A \cap B^c)$
- Are A and B independent? Why?
- Are A and B mutually exclusive? Why

Solution 8.

- $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.2 + (1 - 0.6) - 0.2 = 0.4$
- $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = 0.2/0.4 = 0.5$
- $P(A \cap B^c) = P(A) - P(A \cap B) = 0.2 - 0.2 = 0$
- Since $P(A \cap B) = 0.2 \neq 0.2 \times 0.4 = 0.08$, thus they are not independent.
- No, since $P(A \cap B) \neq 0$

Exercise 9.
Suppose that 0.0003 of all males join criminal groups (CG). Males who join CG are 3 times more likely to deal with drugs compared to those who do not join CGs. What is the probability that a male who deals with drugs has also joins CGs?

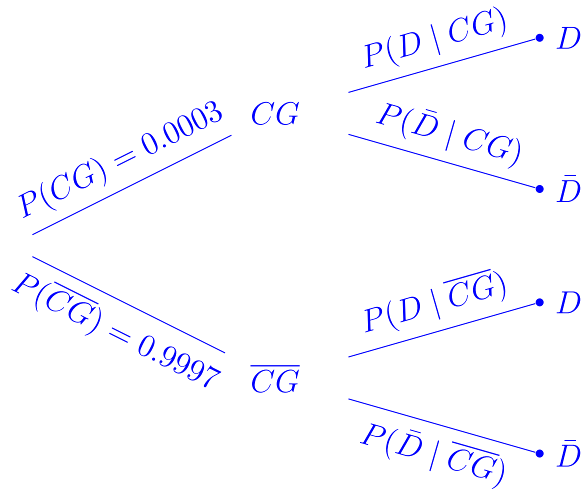
Solution 9.
Define the events

- $CG =$ " Joining criminal groups"
- $\overline{CG} =$ " Not joining criminal groups"
- $D =$ " Dealing with drugs"
- $\bar{D} =$ " Not dealing with drugs"

We know that $P(D \mid CG) = 3P\left(D \mid \overline{CG}\right) = 3p$ We are asked

$$
\begin{aligned}
P(CG \mid D) &= \frac{P(D \mid CG)\, P(CG)}{P(D \mid CG)\, P(CG) + P\left(D \mid \overline{CG}\right) P(\overline{CG})} \\
&= \frac{3 \times \cancel{p} \times (0.0003)}{3 \times \cancel{p} \times (0.0003) + \cancel{p} \times 0.9997} \\
&= \frac{3 \times (0.0003)}{3 \times (0.0003) + \times 0.9997} \\
&\approx 0.0009
\end{aligned}
$$

See the diagram bellow



Exercise 10.
Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats. You pick a door, say No. 1, and the host, who knows what's behind the doors, opens **another** door, say No. 3, which has a goat. He then says to you, "Do you want to pick door No. 2?" Is it to your advantage to switch your choice?

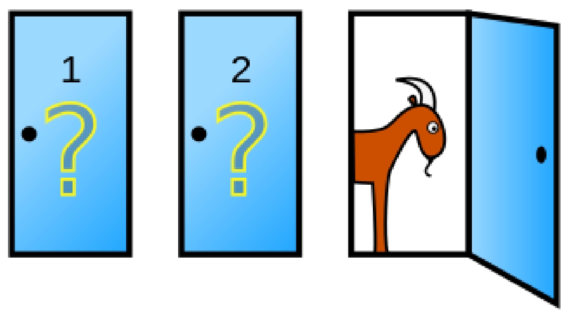


Figure 1: Monty Hall Problem. *From Wikipedia*

Solution 10.

- Let C_i be the event that the car is behind door No. i. We have $\mathbb{P}(C_i) = 1/3$.
- Let H be the event that the host chooses door No. 3. We have,

$$\mathbb{P}(H \mid C_1) = 1/2$$
$$\mathbb{P}(H \mid C_2) = 1$$
$$\mathbb{P}(H \mid C_3) = 0.$$

- By Bayes' rule,

$$\mathbb{P}(C_2 \mid H) = \left(\frac{\mathbb{P}(H \mid C_2)\mathbb{P}(C_2)}{\mathbb{P}(H \mid C_1)\mathbb{P}(C_1) + \mathbb{P}(H \mid C_2)\mathbb{P}(C_2) + \mathbb{P}(H \mid C_3)\mathbb{P}(C_3)}\right)$$
$$= \frac{1}{1 + 1/2} = \frac{2}{3}.$$

It is in your best interests to **switch** doors!

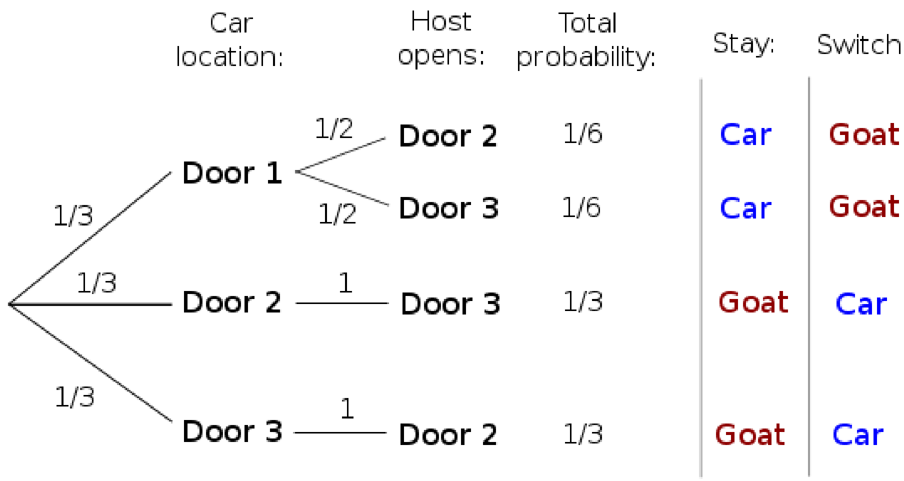


Figure 2: Tree showing the probability of every possible outcome if the player initially picks Door 1: *Taken from wikipedia*